

99.500 Applied Mathematics for Engineering Spring Semester AY2026

Laplace Transform

Week 11 Outline

- Laplace transform for ODE
- Laplace transform for PDE

Heaviside Expansion

To use the Heaviside expansion method for partial fraction decomposition in Laplace Transform, the following three criteria must be met:

1. $P(s)$ and $Q(s)$ must be polynomials of s . ✗ $\sqrt{s} s^{\frac{1}{2}}$ $I_0(\sqrt{s})$
2. The degree of $P(s)$ must be lower than the degree of $Q(s)$ (i.e., $\deg P(s) < \deg Q(s)$).
3. The roots of $Q(s) = 0$ must be distinct (i.e., no repeated roots).

$$Q(s) = (s - \alpha_1)(s - \alpha_2) \cdots (s - \alpha_k)$$

$$\text{then } \mathcal{L}^{-1} \left\{ \frac{P(s)}{Q(s)} \right\} = \sum_{k=1}^N \frac{P(\alpha_k)}{Q'(\alpha_k)} e^{\alpha_k t}$$

$$\bar{f}(s) = \frac{1}{s \cosh \sqrt{s}} = \frac{P(s)}{Q(s)} \quad \begin{matrix} P(s) = 1 \\ Q(s) = s \cosh \sqrt{s} \end{matrix}$$

How about
infinite roots?

$$Q(s) = s \cosh \sqrt{s} = s + \frac{1}{2!} s^2 + \frac{1}{4!} s^3 + \dots$$

$$\cosh x = 1 + \frac{1}{2!} x^2 + \frac{1}{4!} x^4 + \dots$$

$\downarrow$
 $x = \sqrt{s}$

$$1 + \frac{1}{2} s + \frac{1}{24} s^2 + \dots$$

$$\Rightarrow \cosh \sqrt{s} = 0$$

$$\cos(i\sqrt{s})$$

$$\cos(i\sqrt{s}) = 0$$

$$i\sqrt{s} = (k - \frac{1}{2})\pi \quad k = 1, 2, \dots$$

Heaviside Expansion

$$Q(s) = s \left[\cosh \sqrt{s} \right] = 0 \Rightarrow s \left[\cos(i\sqrt{s}) \right] = 0$$

$$\alpha_0 = 0 \text{ \& } i\sqrt{\alpha_k} = (k - \frac{1}{2})\pi, k = 1, 2, 3 \dots \Rightarrow \alpha_k = -(k - \frac{1}{2})^2 \pi^2$$

$$Q = s \cosh \sqrt{s}$$

$$Q'(s) = \cosh \sqrt{s} + \frac{1}{2} \sqrt{s} \sinh \sqrt{s}$$

$$Q'(\alpha_k) = \cosh \sqrt{\alpha_k} + \frac{1}{2} \sqrt{\alpha_k} \sinh \sqrt{\alpha_k}$$

$$= \frac{i}{2} (k - \frac{1}{2}) \pi \sinh[i(k - \frac{1}{2})\pi]$$

$$= \frac{i^2}{2} (k - \frac{1}{2}) \pi \sin[(k - 1/2)\pi]$$

$$= (-1)^k (k - \frac{1}{2}) (\frac{\pi}{2})$$

$$Q'(\alpha_k) = \begin{cases} 1 & (k = 0) \\ \frac{(-1)^k (k - \frac{1}{2}) \pi}{2} & (k > 0) \end{cases}$$

$$-1 \alpha_k = [(k - \frac{1}{2})\pi]^2$$

$$f(t) = \sum_{k=0}^{\infty} \frac{P(\alpha_k)}{Q'(\alpha_k)} e^{\alpha_k t}$$

$$= \frac{1}{1} e^{0t} + \sum_{k=1}^{\infty} \frac{(-1)^k}{(k - \frac{1}{2}) \frac{\pi}{2}} e^{-(k - \frac{1}{2})^2 \pi^2 t}$$

$$= 1 + \frac{2}{\pi} \sum_{k=1}^{\infty} \frac{(-1)^k}{k - \frac{1}{2}} e^{-(k - \frac{1}{2})^2 \pi^2 t}$$

Small-Time Behavior of Laplace Transforms

$$\lim_{t \rightarrow 0} f(t) \sim \mathcal{L}^{-1} \left\{ \lim_{s \rightarrow \infty} \bar{f}(s) \right\}$$

Example:

$$\bar{f}(s) = \frac{s}{s^2 + \omega^2} \xrightarrow{\mathcal{L}^{-1}} f(t) = \cos \omega t$$

Taylor series expansion of $f(t)$:

$$\lim_{t \rightarrow 0} f(t) = 1 - \frac{1}{2!} (\omega t)^2 + \frac{1}{4!} (\omega t)^4 + \underbrace{O((\omega t)^6)}_{+ \frac{1}{6!} (\omega t)^6 + \dots}$$

Small-Time Behavior of Laplace Transforms

Taylor expansion

$$\lim_{s \rightarrow \infty} \bar{f}(s) = \lim_{s \rightarrow \infty} \left[\frac{s}{s^2 + \omega^2} \right] = \lim_{s \rightarrow \infty} \left[\frac{1}{s} \frac{1}{1 + \left(\frac{\omega}{s}\right)^2} \right]$$

$\frac{1 \times s}{\left(s + \frac{\omega^2}{s}\right) \times s} = \frac{s}{s^2 + \omega^2}$

$$\lim_{s \rightarrow \infty} \bar{f}(s) = \frac{1}{s} \left\{ 1 - \left(\frac{\omega}{s}\right)^2 + \left(\frac{\omega}{s}\right)^4 + O\left(\left(\frac{\omega}{s}\right)^6\right) \right\}$$

$\frac{1}{1 + \left(\frac{\omega}{s}\right)^2}$
 $\downarrow$
 small

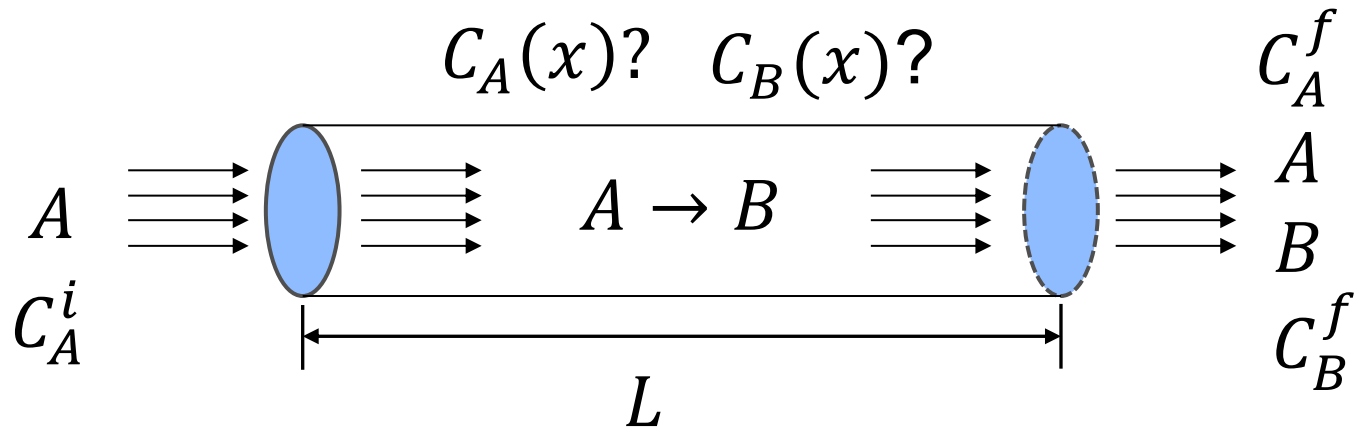
$$\mathcal{L}^{-1} \left[\lim_{s \rightarrow \infty} \bar{f}(s) \right] \sim \mathcal{L}^{-1} \left[\frac{1}{s} - \frac{\omega^2}{s^3} + \frac{\omega^4}{s^5} + O\left(\frac{\omega^6}{s^7}\right) \right]$$

$$\sim 1 - \frac{(\omega t)^2}{2!} + \frac{(\omega t)^4}{4!} + O((\omega t)^6) \sim \lim_{t \rightarrow 0} f(t)$$

$\frac{1}{1+x^2} = 1 - x^2 + x^4 - \dots$
 $x = \frac{\omega}{s}$

Laplace transform tables	
Time space	Laplace
$\frac{t^{n-1}}{(n-1)!}$	$\frac{1}{s^n}$

Example Revisit



1st order reaction: $r_A \rightarrow -kC_A$

$$\frac{dC_A}{dx} = -\frac{k}{u}C_A$$

$$C_A(x=0) = C_{A,0}$$

Let $\alpha = \frac{k}{u}$

$$\mathcal{L}\{\text{ODE}\} \Rightarrow s\bar{C}_A - \underbrace{C_A(0)}_{C_{A,0}} = -\alpha\bar{C}_A$$

$$\Rightarrow s\bar{C}_A - C_{A,0} = -\alpha\bar{C}_A$$

$$\Rightarrow \bar{C}_A = \frac{C_{A,0}}{s + \alpha}$$

“Small-Time” Behavior

$$\bar{C}_A(s) = \frac{C_{A,0}}{s + \alpha} \Rightarrow C_A = L^{-1} \left[\frac{C_{A,0}}{s + \alpha} \right]$$

$$C_A = C_{A,0} e^{-\alpha x} \quad \leftarrow \text{L.T. formula}$$

$$C'_A = -\alpha C_{A,0} e^{-\alpha x}$$

$$e^x = 1 + x + x^2 + x^3 + \dots$$

$$\lim_{x \rightarrow 0} C_A \sim C_{A,0} [1 - \alpha x + O(x^2)]$$

$$\lim_{x \rightarrow 0} C'_A = \lim_{x \rightarrow 0} -\alpha C_{A,0} e^{-\alpha x} \sim -\alpha C_{A,0} [1 + O(x)]$$

$$\frac{1}{1+x} = 1 - x + x^2 - x^3 + \dots$$

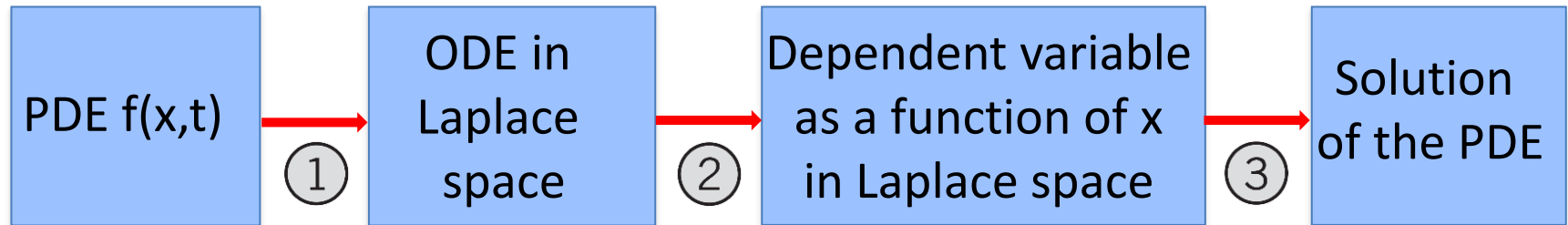
$$\lim_{s \rightarrow \infty} \bar{C}_A = \frac{C_{A,0}}{s + \alpha} = C_{A,0} \left[\frac{1}{s} \frac{1}{1 + \frac{\alpha}{s}} \right] \sim C_{A,0} \left[\frac{1}{s} \left(1 - \frac{\alpha}{s} + O\left(\frac{\alpha^2}{s^2}\right) \right) \right]$$

$$\lim_{s \rightarrow \infty} \bar{C}_A \sim C_{A,0} \left[\frac{1}{s} - \frac{\alpha}{s^2} + O\left(\frac{\alpha^2}{s^3}\right) \right] \quad \leftarrow \text{L.T. formula}$$

$$\lim_{x \rightarrow 0} C_A \sim C_{A,0} [1 - \alpha x + O(x^2)]$$

$$\lim_{x \rightarrow 0} \frac{dC_A}{dx} \sim -\alpha C_{A,0} + O(x)$$

Solving PDE by Laplace Transform



1. Take the Laplace transform with respect to one of the two variables, usually t . This gives an ***ODE for the transform*** of the unknown function. This is so since the derivatives of this function with respect to the other variable slip into the transformed equation. The latter also incorporates the given boundary and initial conditions.
2. Solving that ODE, obtain the transform of the unknown function.
3. Taking the inverse transform, obtain the solution of the given problem.

Heat Conduction Example 1

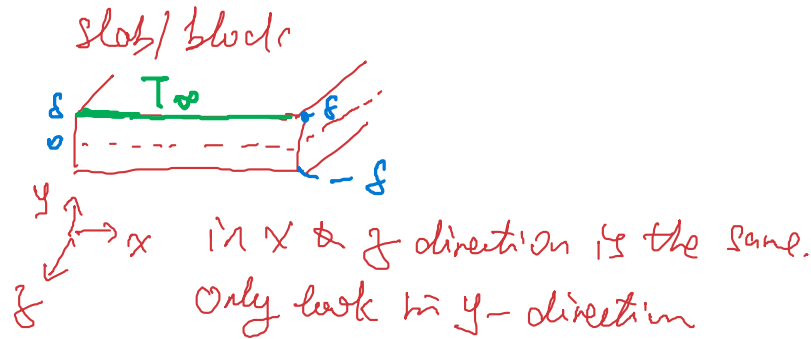
$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial y^2}$$

$$T(y, 0) = T_0 \quad \text{IC}$$

$$T(\delta, t) = T_\infty \quad \text{BC1}$$

$$\left. \frac{\partial T}{\partial y} \right|_{y=0} = 0 \quad \text{BC2.}$$

symmetric



Transform the PDE and BCs into the Laplace space:

$$\mathcal{L} \left\{ \frac{\partial T}{\partial t} \right\} = \mathcal{L} \left\{ \alpha \frac{\partial^2 T}{\partial y^2} \right\} \Rightarrow s\bar{T} - T(y, 0) = \alpha \frac{d^2 \bar{T}}{dy^2} \quad \text{Transformed PDE}$$

$$\text{BC1: } \bar{T}(\delta) = \frac{T_\infty}{s}$$

$$\text{BC2: } \left. \frac{d\bar{T}}{dy} \right|_{y=0} = 0$$

L.T.
formula

Heat Conduction Example 1

2nd order ODE.

$$\lambda^2 - \frac{s}{\alpha} = 0 \Rightarrow \frac{d^2 \bar{T}}{dy^2} - \frac{s}{\alpha} \bar{T} = -\frac{T_0}{\alpha}$$

$$\lambda = \sqrt{\frac{s}{\alpha}} \Rightarrow \bar{T}_h = c_1 \cosh(y \sqrt{\frac{s}{\alpha}}) + c_2 \sinh(y \sqrt{\frac{s}{\alpha}})$$

$$\Rightarrow \bar{T}_p = \frac{T_0}{s} \text{ particular solution}$$

$$\Rightarrow \bar{T} = c_1 \cosh(y \sqrt{\frac{s}{\alpha}}) + c_2 \sinh(y \sqrt{\frac{s}{\alpha}}) + \frac{T_0}{s}$$

$c_2 = 0$
↑
to be 0

$$\bar{T}(\delta) = \frac{T_\infty}{s}$$

$$\bar{T} = c_1 \cosh(y \sqrt{\frac{s}{\alpha}}) + \frac{T_0}{s}$$

$$\text{BC2} \Rightarrow c_2 = 0$$

$$\bar{T}(\delta) = c_1 \cosh(\delta \sqrt{\frac{s}{\alpha}}) + \frac{T_0}{s}$$

$$\text{BC1} \Rightarrow c_1 = \frac{T_\infty - T_0}{s} \frac{1}{\cosh(\delta \sqrt{\frac{s}{\alpha}})}$$

$$y=0 \Rightarrow \frac{d\bar{T}}{dy} = \cancel{\sinh(y \sqrt{\frac{s}{\alpha}})} + \cancel{\frac{1}{\delta} \cosh(y \sqrt{\frac{s}{\alpha}})} = 0$$

$$\bar{T}(y) = \frac{T_0}{s} + \frac{T_\infty - T_0}{s} \frac{\cosh(y \sqrt{\frac{s}{\alpha}})}{\cosh(\delta \sqrt{\frac{s}{\alpha}})} = \frac{T_0}{s} + \frac{T_\infty - T_0}{s} \frac{\cos(iy \sqrt{\frac{s}{\alpha}})}{\cos(i\delta \sqrt{\frac{s}{\alpha}})}$$

$$T(y, t) = \mathcal{L}^{-1}\{\bar{T}(y)\} \text{ Heaviside?}$$

$$T(y, t) = T_0 + (T_\infty - T_0) \mathcal{L}^{-1} \left\{ \frac{P(s)}{Q(s)} \right\} \quad \begin{aligned} P(s) &= \cos(iy \sqrt{\frac{s}{\alpha}}) \\ Q(s) &= s \cos(i\delta \sqrt{\frac{s}{\alpha}}) \end{aligned}$$

$s=0$
 $\Rightarrow 0 \cdot \cos(i\delta \sqrt{\frac{s}{\alpha}}) = 0$

Heat Conduction Example 1

Check 1: $P(s), Q(s)$ polynomials in s ? (Yes)

Check 2: Order of $P(s) <$ Order of $Q(s)$? (Yes)

Check 3: Distinct roots? Roots are: $s_0 = 0, i\sqrt{\frac{s_n}{\alpha}}\delta = (n - 1/2)\pi, n = 1, 2, \dots$

$$\cos(i\delta\sqrt{\frac{s}{\alpha}}) = 0$$

$\Downarrow$

$$i\delta\sqrt{\frac{s_n}{\alpha}} = (n - \frac{1}{2})\pi$$

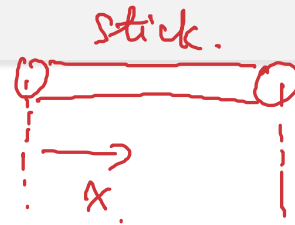
$$P(s_n) = \cos\left(i\delta\sqrt{\frac{s_n}{\alpha}}\right) = \cos\left(\frac{(n - \frac{1}{2})\pi}{i\delta}\right)$$

$$T(y, t) = T_0 + (T_\infty - T_0) \left(\sum_{n=0}^{\infty} \frac{P(s_n)}{Q'(s_n)} e^{s_n t} \right)$$

$$= T_0 + \underbrace{(T_\infty - T_0) \cdot \frac{1}{1} e^0}_{s=0} + (T_\infty - T_0) \underbrace{\sum_{n=1}^{\infty} \frac{\cos\left[(n - \frac{1}{2})\pi \frac{y}{\delta}\right]}{(-1)^n (n - \frac{1}{2}) \frac{\pi}{2}} e^{-(n - \frac{1}{2})^2 \pi^2 \frac{\alpha t}{\delta^2}}}_{n \geq 1}$$

$$T(y, t) = T_\infty + \frac{2}{\pi} (T_\infty - T_0) \sum_{n=1}^{\infty} \frac{(-1)^n \cos\left[(n - \frac{1}{2})\pi \frac{y}{\delta}\right]}{(n - \frac{1}{2})} e^{-(n - \frac{1}{2})^2 \pi^2 \frac{\alpha t}{\delta^2}}$$

Heat Conduction Example 2



Nondimensionalization:

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}$$

$$\text{let } \theta = \frac{T - T_L}{T_0 - T_L} \Rightarrow \frac{\partial \theta}{\partial t} = \alpha \frac{\partial^2 \theta}{\partial x^2}$$

$$x=0 \quad T_0$$

$$x=L \quad T_L$$

$$\text{BC1 } T(0, t) = T_0$$

$$\text{BC2 } T(L, t) = T_L$$

$$T(x, 0) = T_L \quad \text{IC}$$

$$\theta(0, t) = 1$$

$$\theta(L, t) = 0$$

$$\theta(x, 0) = 0$$

$$\text{Apply BC1: } C_1 \cosh 0 + C_2 \sinh 0 = \frac{1}{s} \Rightarrow C_1 = \frac{1}{s}$$

$$\text{BC2: } \frac{1}{s} \cosh(L\sqrt{\frac{s}{\alpha}}) + C_2 \sinh(L\sqrt{\frac{s}{\alpha}}) = 0$$

$$C_2 = -\frac{\cosh(L\sqrt{\frac{s}{\alpha}})}{s \sinh(L\sqrt{\frac{s}{\alpha}})}$$

Transform the PDE and BCs into the Laplace space:

$$\mathcal{L} \left\{ \frac{\partial \theta}{\partial t} \right\} = \mathcal{L} \left\{ \alpha \frac{\partial^2 \theta}{\partial x^2} \right\} \Rightarrow s\bar{\theta} - \theta(x, 0) = \alpha \frac{d^2 \bar{\theta}}{dx^2}$$

$$\text{BC1: } \bar{\theta}(0) = \frac{1}{s}$$

$$\text{BC2: } \bar{\theta}(L) = 0$$

$$\text{2nd order ODE: } \frac{d^2 \bar{\theta}}{dx^2} - \frac{s}{\alpha} \bar{\theta} = 0 \Rightarrow \bar{\theta} = c_1 \cosh(x\sqrt{\frac{s}{\alpha}}) + c_2 \sinh(x\sqrt{\frac{s}{\alpha}})$$

$$\frac{1}{s} \cosh(x\sqrt{\frac{s}{\alpha}}) - \frac{\cosh(L\sqrt{\frac{s}{\alpha}})}{s \sinh(L\sqrt{\frac{s}{\alpha}})} \sinh(x\sqrt{\frac{s}{\alpha}})$$

$$\sinh\left[(L-x)\sqrt{\frac{s}{\alpha}}\right] = \frac{1}{s} \left[\cosh(x\sqrt{\frac{s}{\alpha}}) \sinh(L\sqrt{\frac{s}{\alpha}}) - \cosh(L\sqrt{\frac{s}{\alpha}}) \sinh(x\sqrt{\frac{s}{\alpha}}) \right]$$

Restricted

Heat Conduction Example 2

$$e^x \sim 1 + x + \frac{x^2}{2!} + \dots$$

$$\sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots$$

$$\bar{\theta} = \frac{\sinh[(L-x)\sqrt{\frac{s}{\alpha}}]}{s \sinh(L\sqrt{\frac{s}{\alpha}})}$$

$$= \frac{1}{s} \left[\frac{\left(\frac{L-x}{\sqrt{\alpha}}\sqrt{s} + \left(\frac{L-x}{\sqrt{\alpha}}\right)^3 \frac{s^{\frac{3}{2}}}{3!} + \left(\frac{L-x}{\sqrt{\alpha}}\right)^5 \frac{s^{\frac{5}{2}}}{5!} + O(s^{\frac{7}{2}})}{\frac{L}{\sqrt{\alpha}}\sqrt{s} + \left(\frac{L}{\sqrt{\alpha}}\right)^3 \frac{s^{\frac{3}{2}}}{3!} + \left(\frac{L}{\sqrt{\alpha}}\right)^5 \frac{s^{\frac{5}{2}}}{5!} + O(s^{\frac{7}{2}})} \right]$$

$$= \frac{\frac{L-x}{\sqrt{\alpha}} + \left(\frac{L-x}{\sqrt{\alpha}}\right)^3 \frac{s}{3!} + \left(\frac{L-x}{\sqrt{\alpha}}\right)^5 \frac{s^2}{5!} + O(s^3)}{\frac{L}{\sqrt{\alpha}}s + \left(\frac{L}{\sqrt{\alpha}}\right)^3 \frac{s^2}{3!} + \left(\frac{L}{\sqrt{\alpha}}\right)^5 \frac{s^3}{5!} + O(s^4)} = \frac{P(s)}{Q(s)} = \frac{\frac{\sinh[(L-x)\sqrt{\frac{s}{\alpha}}]}{\sqrt{s}}}{\sqrt{s} \sinh(L\sqrt{\frac{s}{\alpha}})}$$

$$P(s) = \frac{\sinh[(L-x)\sqrt{\frac{s}{\alpha}}]}{\sqrt{s}} \Rightarrow s = 0 \text{ or } \sinh[L\sqrt{\frac{s}{\alpha}}] = -i \sin[iL\sqrt{\frac{s}{\alpha}}]$$

$$\Rightarrow \left(iL\sqrt{\frac{s_n}{\alpha}}\right)^2 = (n\pi)^2, n = 1, 2, 3, \dots$$

$$Q(s) = \sqrt{s} \sinh(L\sqrt{\frac{s}{\alpha}}) = 0 \Rightarrow s_n = -\alpha \left(\frac{n\pi}{L}\right)^2, n = 1, 2, 3, \dots$$

$$Q'(s) = \frac{1}{2}s^{-\frac{1}{2}} \sinh[L\sqrt{\frac{s}{\alpha}}] + \frac{1}{2} \frac{L}{\sqrt{\alpha}} \cosh[L\sqrt{\frac{s}{\alpha}}]$$

Heat Conduction Example 2

$$\cosh x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \dots$$

$\lim_{s \rightarrow 0} \left[\frac{P(s)}{Q'(s)} \right] = \frac{\frac{L-x}{\sqrt{\alpha}} + O(s)}{\frac{1}{2} \frac{L}{\sqrt{\alpha}} + O(s) + \frac{1}{2} \frac{L}{\sqrt{\alpha}} [1 + O(s)]} = \frac{L-x}{L} = 1 - \frac{x}{L}$

$\int \sqrt{\frac{s_n}{\alpha}} = n\pi$
 $\int \sqrt{\frac{s_n}{\alpha}} = \frac{n\pi}{i} = -i n\pi$

$\theta(x, t) = 1 - \frac{x}{L} + \sum_{n=1}^{\infty} \frac{P(s_n)}{Q'(s_n)} e^{s_n t}$

$Q'(s_n) = 0 + \frac{1}{2} \frac{L}{\sqrt{\alpha}} \cosh(-in\pi) = \frac{1}{2} \frac{L}{\sqrt{\alpha}} \cosh(-in\pi) = \frac{1}{2} \frac{L}{\sqrt{\alpha}} \cos(n\pi)$

$P(s_n) = \frac{\sinh[(L-x)(\frac{in\pi}{L})]}{\sqrt{s} = \frac{i\sqrt{\alpha}n\pi}{L}} = \frac{\sinh[(L-x)(\frac{in\pi}{L})]}{\frac{i\sqrt{\alpha}n\pi}{L}} = \frac{i \sin[(1 - \frac{x}{L})n\pi]}{\frac{i\sqrt{\alpha}n\pi}{L}}$

$\Rightarrow \frac{P(s_n)}{Q'(s_n)} = \frac{\frac{\sin[(1 - \frac{x}{L})n\pi]}{\frac{\sqrt{\alpha}n\pi}{L}}}{\frac{(-1)^n L}{(2\sqrt{\alpha})}} = \frac{(-1)^n}{n\pi} 2 \sin \left[\left(1 - \frac{x}{L}\right) n\pi \right]$

$\Rightarrow \theta(x, t) = 1 - \frac{x}{L} + \sum_{n=1}^{\infty} (-1)^n \frac{2 \sin \left[\left(1 - \frac{x}{L}\right) n\pi \right]}{n\pi} e^{-\alpha \left(\frac{n\pi}{L}\right)^2 t}$

$\cos(i(-i)n\pi) = \cos(n\pi)$

$\cosh(-in\pi) = \cos(n\pi)$

$\sinh[(L-x)(\frac{in\pi}{L})] = i \sin[(1 - \frac{x}{L})n\pi]$

Announcements

Week 11 Homework

- Due Date: Week 12 class – 2:00 PM, April 15, 2026

Week 12 Lab Session

- Lucky draw for computer test grouping in week 13

Week 13 Computer Test & Exam Instruction

Week 14 Individual Meetings

- Date: Tuesday, April 28, 2026
- Time: 10:00 AM – 4:00 PM
- Action Required: Please indicate your time slot using this link

<https://docs.google.com/spreadsheets/d/10sE6iWVvFoSE9Oo5DYvNaOhwu1nqeNufKXq8NkKbHRs/edit?gid=0#gid=0>